

TABLE OF CONTENT:

CHAPTER NO:	TITLE:	PAGE NO:
	Abstract	iii
	Acknowledgement	iv
	List of Tables	vi
	List of Figures	vii
	List of Abbreviations	viii
1	CHAPTER 1	
1.1	Introduction	
1.2	Educational Landscape in Tamil Nadu	
1.3	Challenges in the Current Admission System	
1.4	Role of Technology in Modern Education	
1.5	Motivation Behind the Project	
1.6	Project Vision and Scope	
1.7	Target Users	
1.8	Expected Impact	
1.9	Innovation and Uniqueness	
1.10	Organization of the Report	
2	CHAPTER 2 - PROBLEM STATEMENT:	
2.1	Context and Background	
2.2	Current Challenges and Pain Points	

2.2.1	Information Fragmentation	
2.2.2	Unpredictability of Cutoffs	
2.2.3	Lack of Personalized Guidance	
2.2.4	Limited Scholarship Awareness	
2.3	Problem Magnitude and Urgency	
2.4	Existing Solutions and Limitations	
2.5	Gap Analysis	
2.6	Problem Statement Summary	
2.7	Research Questions	
2.8	Proposed Solution Approach	
2.9	Expected Outcomes	
3	CHAPTER 3: OBJECTIVES	
3.1	Primary Objectives	
3.1.1	Accurate Prediction System	
3.1.2	Centralized Information Repository	
3.1.3	Personalized Guidance	
3.2	Secondary Objectives	
3.2.1	Accessibility and User-Friendliness	
3.2.2	Data Security	
3.3	Technical Objectives	
3.4	Functional Objectives	
3.5	Long-Term Objectives	
3.6	Success Criteria	
4	CHAPTER 4: SCOPE OF THE PROJECT	
4.1	Introduction to Project Scope	
4.2	Functional Scope	

4.2.1	Admission Prediction	
4.2.2	College Database	
4.2.3	Scholarship Matching	
4.2.4	Scholarship Matching	
4.3	Technical Scope	
4.4	Data Scope	
4.5	User Scope	
4.6	Geographic Scope	
4.7	Time Scope	
4.8	Limitations and Constraints	
5	CHAPTER 5: LITERATURE REVIEW	
5.1	Introduction to Literature Review	
5.2	Machine Learning in Education	
5.3	Survey of Existing Admission Systems	
5.4	Classification Algorithms (Random Forest, etc.)	
5.5	Regression Models for Cutoff Prediction	
5.6	NLP in Education (Chatbots)	
5.7	Comparative Analysis of Existing Systems	
5.8	Research Gaps Identified	
6	CHAPTER 6: SYSTEM ANALYSIS	
6.1	Requirements Analysis	
6.1.1	Functional Requirements	
6.1.2	Non-Functional Requirements	
6.2	Feasibility Study	

6.2.1	Technical Feasibility	
6.2.2	Economic Feasibility	
6.3	System Modeling	
6.3.1	Use Case Diagrams	
6.3.2	Data Flow Diagrams (DFD)	
6.4	Data Modeling (ER Diagrams)	
7	CHAPTER 7: SYSTEM DESIGN	
7.1	System Architecture (MVC)	
7.2	Database Design	
7.2.1	Schema and Tables	
7.2.2	Relationships	
7.3	Machine Learning Model Design	
7.3.1	Preprocessing Pipeline	
7.3.2	Model Selection	
7.4	User Interface Design	
7.5	User Interface Design	
7.6	Security Design	
8	CHAPTER 8: IMPLEMENTATION	
8.1	Implementation Environment (Tools & Tech Stack)	
8.2	Database Implementation (SQLite)	
8.3	Machine Learning Implementation (Scikit-learn)	
8.4	Backend Implementation (Flask)	
8.5	Frontend Implementation (HTML/Bootstrap)	

8.6	Chatbot Implementation	
8.7	Code Snippets and Key Functions	
9	CHAPTER 9: TESTING	
9.1	Testing Strategy	
9.2	Unit Testing	
9.3	Integration Testing	
9.4	System Testing	
9.5	Performance Testing	
9.6	Security Testing	
9.7	Test Cases and Results	
10	CHAPTER 10: RESULTS AND DISCUSSION	
10.1	System Functionality Demonstration	
10.2	Model Performance (Accuracy & Metrics)	
10.3	Prediction Accuracy Analysis	
10.4	System Performance (Response Time)	
10.5	User Interface Evaluation	
10.6	Comparative Analysis	
10.7	Limitations Observed	
11	CHAPTER 11: CONCLUSION AND FUTURE WORK	
11.1	Summary of the Project	
11.2	Key Achievements	
11.3	Objectives Fulfilled	

11.4	Challenges Overcome	
11.5	Future Enhancements	